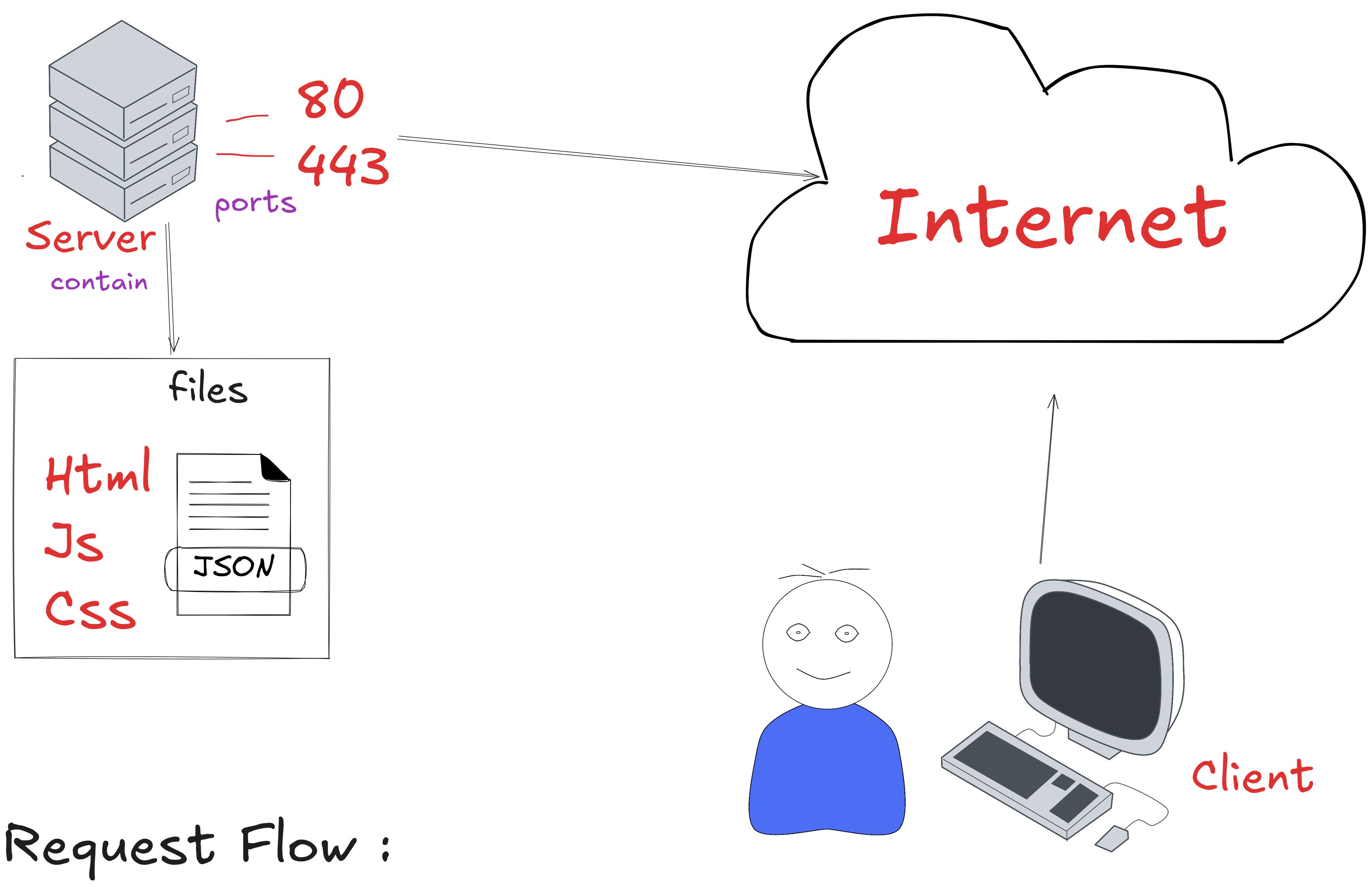
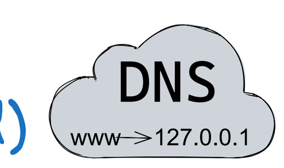
What is The backend and How they Work?



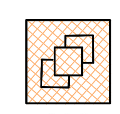
Request Flow :

1. DNS Server (The browser sends the domain name to a DNS Server, which translates it into an IP address using an A Record)

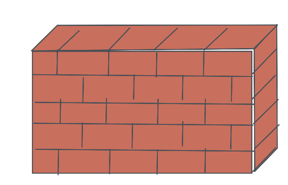


2. AWS (EC2) The request travels over the internet to that specific IP address, landing at a cloud instance, such as an AWS EC2 server

EC2 = Elastic Compute Cloud



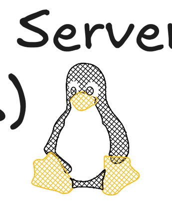
3. Firewall (Security Groups) Before entering the operating system, the request must pass through a firewall. It verifies if the incoming ports are allowed
   protocols (Tcp/Udp)
   Scr/Des
   Ip(Version)



Firewall

4.Nginx A Reverse proxy like Nginx is a server which sets IN front the server . It receives the request, so we can Manage different types of redirects or configs from a centralized Space

5. Local Server (Node.js) (The final hop. Nginx routes the incoming request internally to a local port like Localhost:3001. This is where your actual backend code executes, processes the data, and sends the response back)



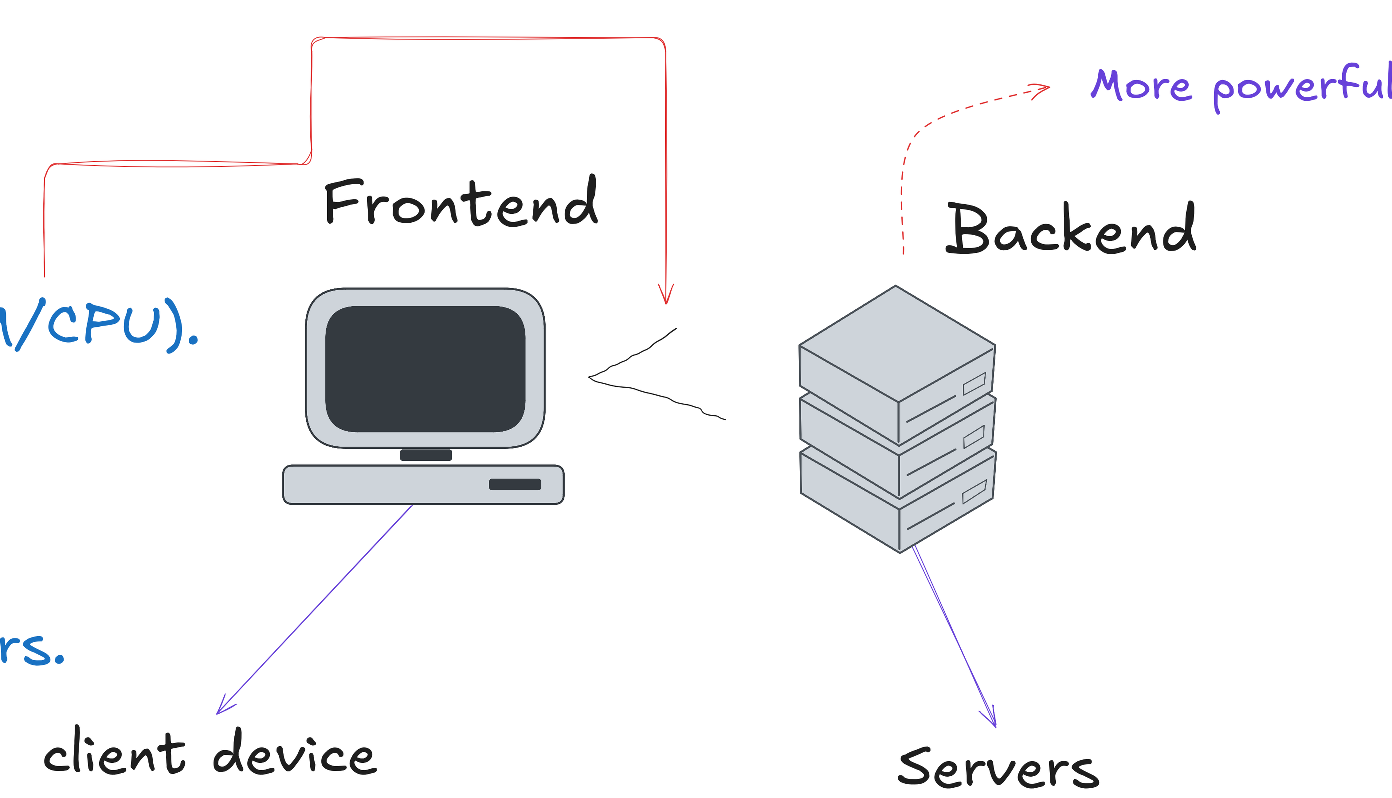
Why NOT write Backend logic in Frontend ?

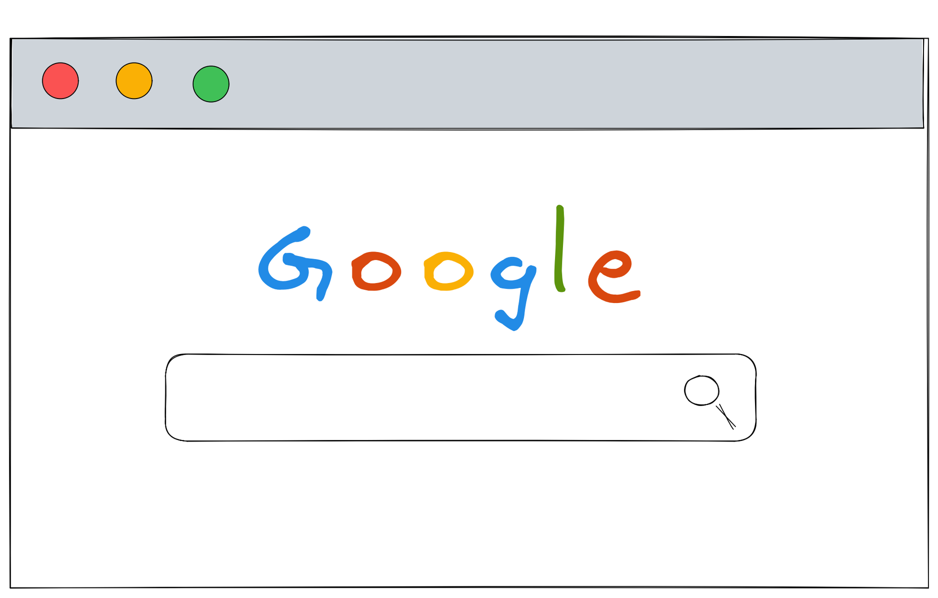
Security Restrictions (Browsers are sandboxed environments. They restrict direct access to the underlying OS, file systems, and external APIs due to CORS policies to protect the user's machine)

Database Connections Browsers are not designed to maintain persistent socket connections and would overwhelm the DB server

Computing Power (Frontend code runs on the Client's device (which might have limited RAM/CPU). Backend logic runs on centralized, scalable servers that can handle heavy processing)

Centralized State (A Backend acts as a Centralized Computer that holds data for ALL users. This is necessary for interactions between users, like User A sending a notification to User B)





Web Application

Frontend Runtime (The code is fetched from the server, but it is EXECUTED by the Browser locally on the client's machine)

Backend Runtime (The request is sent to the server, and the actual processing and logic happen ON the Server itself)